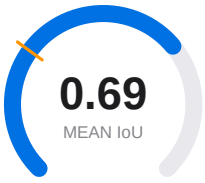
What this document is not. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.

FINDING

419420729 is the best physical fit.

Forward replay tested 4 candidates; the runner-up fits at 0.54. A ranking by physical consistency under tested assumptions — an investigative lead for a human analyst.



MEAN IOU	VARIANTS WON	LEAD RETAINED	CANDIDATES
0.69	**5** / 6	**82%**	**4**
the fit gate is 0.3	forcing variants where it fits best	of 80 tested changes	replayed by forward replay

EVIDENCE

- Best fit under **5 of 6** forcing variants
- Ahead of the runner-up by a mean loss gap of **0.1071**
- Inside the origin area **24 h** before the image
- The nearest vessel, **419623584**, 0.0036° away, is not supported

Candidates, ranked by physical fit

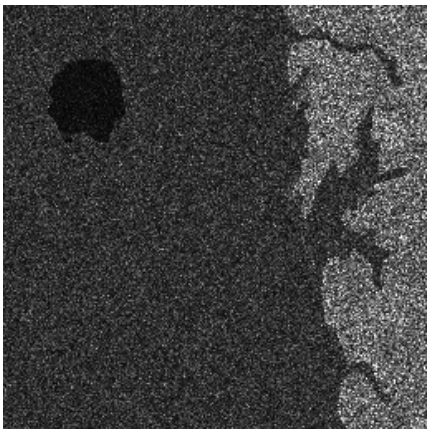
Each candidate's release replayed forward under six forcing variants and compared with the observed slick.

#	VESSEL	MEAN IoU		VARIANTS WON	RELEASE STARTS
1	419420729		0.686	5 of 6	9 Sep 2026, 22:30 UTC
2	419956187		0.542	1 of 6	10 Sep 2026, 02:00 UTC
3	419623584		0.210	0 of 6	10 Sep 2026, 03:30 UTC
4	419325997		0.051	0 of 6	9 Sep 2026, 22:30 UTC

The proximity baseline — the vessel nearest the slick when the image was taken — would pick **419623584**, ranked 3 by forward replay.

The scene

The radar image the slick was detected in. Radar darkness alone is not proof of oil.



Acquired	10 Sep 2026, 04:30 UTC
Sensor	synthetic SAR-like (VV proxy, single channel)
Pixel spacing	433.59 m
Data status	synthetic
Label source	generator ground truth (synthetic) — not a human or independently confirmed label
Projection	EPSG:32643

Robustness

Does the leader survive small changes to the model's assumptions?

82%
LEAD RETAINED

80 tested changes to the model's assumptions; **14** changed the leader.

The smallest change that flips it: **diffusivity multiplier 1.4** (magnitude 0.5), under which **419956187** leads.

Single-axis and two-axis perturbations only. Three-or-more-axis combinations were not tested, so absence of a flip is evidence of stability across the tested set, not a proof of stability.

Forward exposure

Where the oil is modelled to drift after the image. Reanalysis-style retrospective field — not an operational forecast.

AFTER THE IMAGE	MODELLED SPREAD	SHORELINE	FISHING GROUND	MANGROVE RESERVE
+6 h 10 Sep 2026, 10:30 UTC	479 km^2	0%	42%	0%
+12 h 10 Sep 2026, 16:30 UTC	536 km^2	0%	40%	0%
+24 h 11 Sep 2026, 04:30 UTC	582 km^2	0%	29%	0%

Traffic filtering

Every vessel in the scene, checked against where the oil could have come from over the 24 hours before the image.

4	Candidate	track data present and inside the backward envelope at some queried time
1	Unassessed	never inside where track data exists, but no track data at one or more queried times — missing evidence, NOT exoneration
1	Excluded	track data at every queried time, never inside the envelope

Unassessed is reported separately from excluded: missing evidence is not exoneration.

Coverage-aware matching

Matching radar contacts to AIS when some AIS is hidden — tested on synthetic contacts.

RATE	VESSELS HIDDEN	AREA GOES DARK
Hidden AIS recovered	**79%** 50/63	**83%** 75/90
False alerts	**27%** 31/113	**17%** 20/119
Mistaken promotions	**18%** 7/40	**80%** 24/30

Provenance

AIS	synthetic (fictional MMSIs; no real broadcast data)
Forcing	slicktrace_core.synth_generator constant+spatial field v0.2
Licence	Synthetic data generated by this repository; released under the repository licence. No third-party data terms apply.
Generated	26 Sep 2026, 17:11 UTC
Software	Python 3.12.14 · Darwin 25.2.0 (arm64)

File integrity

Hashes verify file integrity only. They do not establish that an observation is true, that a label is correct, or that this analysis is legally sufficient.

attribution_result.json	`1ce44b82fc967942b33ba2c032f02bba60c7a3f6a0697159a35f87f3d55c042b`
coverage_result.json	`e16cb40e5a418deedf9b9b52fe26f7c6f6581acd03238354f306fdd43cc4e1c2`
forcing.npz	`34b71c9eccb0e5c6e6040bb3d693733b738a1d2bb059f81a7f790e734cffb928`
forecast_result.json	`f9decb564b65b8d2a3643974a13e374e63a1ea15814234be5a089cda3cd1a372`
fragility_result.json	`12669ef777c216119ba79a5d52d74d35a18daa20bb525569d90c43da66bf1c2f`
historical_mapping.json	`927931707c12bdd44a5917c83ae7b6cf09d1939fe0c0bec8e7570bc9b604e6f7`
manifest.json	`102109c76fc452c77b311c92a9b3032e6d17af60608f882b80b8f53be231ab86`
mask_groundtruth.png	`0ea1184f6549eef2259bb1e0133e13ced5fffd808ba3517d47e395b50cf37441`
naive_baseline_result.json	`470ff4ec0ba05f46a41403072c86050f8106dcef7b670572b19e257e38745b8a`
next_observation_result.json	`6f6f1e25763cbd90c2da429e3f58bb46e009d5816d0e9c5cd72645280abd04af`
scene.png	`39be2f9e4d7a757f70d852738ed47c8df04620c44fd735cd7bb4cac6e0f3d950`

Uncertainty

— Rankings express physical consistency under tested assumptions. They are not proof of responsibility.

— Ensemble and perturbation-grid win frequencies are evidence scores, not calibrated probabilities. No calibration against held-out real outcomes exists.

— Where no candidate clears the absolute fit gate the system reports unknown source rather than naming the least-bad option.

— Synthetic ranking accuracy is not a measurement of real-world attribution accuracy.

Restated. This is not a court-ready dossier, a legal finding, or an accusation. It records what the system computed and under which assumptions.